



FPOWS

Field Protection Operations Web System
PROJECT PLAN

simPRO Automation Platform · REDMEN Fire Protection

Delivery History · Current State · Roadmap

Bryan Morales · First Deployed 17 March 2026

ABOUT

FPOWS is an internal web platform built to automate and streamline field operations for REDMEN Fire Protection. It connects directly to simPRO — the company's field service management system — and provides technicians and managers with real-time job data, automated reporting, and digital call sheets, eliminating manual lookups and paper-based workflows.

OBJECTIVES

- ▶ Centralise access to simPRO job data through a fast, purpose-built web UI
- ▶ Automate generation and delivery of job call sheet emails to clients
- ▶ Auto-sync asset next-service dates back to simPRO after job completion
- ▶ Give management a daily live portfolio summary without manual effort
- ▶ Track live technician presence across active jobs
- ▶ Provide a secure, hardened admin interface for log management

STAKEHOLDERS

Project Owner	Bryan Morales
Primary Users	REDMEN Fire Protection technicians & admin
Report Recipient	MANAGER_EMAIL (daily portfolio summary)
Platform	Google Cloud Run (hosted, always-on)
Source System	simPRO (jobs, customers, assets)

PROJECT METRICS

7
DELIVERY PHASES

51
GIT COMMITS

63
DAYS IN DEV

17
API ENDPOINTS

TECH SUMMARY

Runtime	Node.js (ESM) · Express 5
Deployment	Google Cloud Run · deploy.ps1
Scheduling	node-cron (06:00 Asia/Manila daily)
Email	Gmail SMTP · nodemailer · pool(3)
Security	helmet · rate-limit · ADMIN_API_KEY
Current Version	v1.2.1
First Deployed	17 March 2026
Last Updated	21 April 2026 (feat) · 18 May 2026 (security)

SCOPE

- ▶ simPRO REST API integration (read + write)
- ▶ Web UI — job viewer, customer search, job detail modal
- ▶ Email call sheets — HTML emails with logo + history log
- ▶ Daily & bi-weekly manager portfolio reports
- ▶ Asset service date auto-sync (6-monthly / 12-monthly)
- ▶ Live presence tracking per job (in-memory)
- ▶ Admin panel — logs, history management, manual triggers
- ⚠ Out of scope: GoFormz integration (planned, not started)
- ⚠ Out of scope: Per-user authentication (currently API key only)

PHASES 1 – 4

1

17 March 2026 · 08:37 – 12:22

Foundation & Cloud Run Deployment

Initial simPRO integration

Cloud Run env var fixes

URL sanitization

Dockerfile

Axios baseURL fix

Production-ready deployment

2

17 – 18 March 2026

Core UI & Chatbot Widget

FPOWS Agent Widget sidebar

Chatbot Brain 2.0

Context tracking

Negation handling

Job dates in dropdown

Typing latency

3

20 – 26 March 2026

Production v1.2.1 & Email Call Sheets

Manual save workflow

Editable metadata

Email call sheet delivery

Manager email fix

Control bar layout

Email testing intercept

Lead # column

4

26 – 27 March 2026

Email History & Branding

Client message in email template

Sent history log view

History deletion

Branding updates

Chatbot tone refinement

PHASES 5 – 7

5

1 – 9 April 2026

Smart Customer Search & Presence

Priority customer dropdown

60s in-memory cache

Batch enrichment (3×300ms)

Overdue detection (30 days)

customerAssets service dates

Live presence tracking

Print layout

Site-scoped works

6

20 – 21 April 2026

Reporting & Asset Sync

Bi-weekly master summary email

Portfolio site discovery

Live per-site data fetch

Asset sync optimisation

Overdue logic refinement

Email config fix

7

18 May 2026

Security Hardening

ADMIN_API_KEY authentication

helmet (HTTP headers)

express-rate-limit (3 limiters)

Credentials moved to .env

fs.promises async I/O

window.FPOWS_KEY injection

/health endpoint

Development Pace: The majority of core features were built and deployed within the first 2 weeks (17–27 March 2026). Subsequent phases focused on intelligence, search quality, and reporting. The project reached production-stable status at v1.2.1 on 25 March 2026.

LIVE FEATURES

WEB UI

- ✓ Sticky glassmorphic control bar (brand, search, actions)
- ✓ Today's job / schedule viewer from simPRO
- ✓ Customer search — smart cache with batch enrichment
- ✓ Priority (★ Major) customer tagging via simPRO tags
- ✓ Overdue customer detection (30-day threshold)
- ✓ Job detail modal — site, service dates, outstanding works
- ✓ Job description cleaning (strips 30+ boilerplate patterns)
- ✓ Job history per customer (last 30, stage-sorted)
- ✓ Live presence — who's viewing each job (15s heartbeat)
- ✓ Print layout for job call sheets

EMAIL & REPORTING

- ✓ HTML email call sheets — to client + CC manager
- ✓ CID logo attachment (logo.png)
- ✓ Email history log (email_history.jsonl)
- ✓ Admin log viewer + per-entry deletion
- ✓ Daily portfolio report — 06:00 Asia/Manila via node-cron
- ✓ Bi-weekly master summary — on demand / Cloud Scheduler
- ✓ Report content: per-site service dates + outstanding works

SIMPRO INTEGRATION

- ✓ Read: jobs, customers, sites, assets, schedules
- ✓ Write: PATCH asset service dates (6mo / 12mo sync)
- ✓ Rate limit handling — 429 exponential backoff (3 retries)
- ✓ URL validation + 15s timeout per request
- ✓ Company ID configurable via env var

KNOWN LIMITATIONS

- ⚠ Customer search cache can be slow to build (250 jobs + batch enrichment)
- ⚠ node-cron re-registers on every Cloud Run cold start — may miss 06:00 if instance is scaled to zero
- ⚠ Presence tracking is in-memory only — lost on container restart
- ⚠ No per-user login — all users share the same ADMIN_API_KEY
- ⚠ email_history.jsonl is ephemeral on Cloud Run (no persistent volume)
- ⚠ No webhook from simPRO — asset sync must be triggered manually
- ⚠ Chatbot widget (Brain 2.0) deployed but not prominently surfaced in current UI

SECURITY STATUS

HTTP headers	✓ helmet (secured)
Rate limiting	✓ 3 limiters (search, email, admin)
Admin auth	✓ ADMIN_API_KEY (x-admin-key header)
Credentials	✓ .env (gitignored, Cloud Run env vars)
Input validation	✓ Zod (MCP) · integer range check (job IDs)
URL safety	✓ new URL() validation before every API call
Per-user auth	✗ Not implemented
HTTPS	✓ Enforced by Cloud Run (TLS termination)

DEPLOYMENT STATUS

Platform	Google Cloud Run
Version	v1.2.1
Deploy script	deploy.ps1 (PowerShell)
Health check	GET /health — checks 4 required env vars
Cron timezone	Asia/Manila (UTC+8)
Last commit	21 April 2026 (feat) · 18 May 2026 (security)

SHORT TERM

Persistent Email History

Planned

- ▶ Move email_history.jsonl to Cloud Storage or a persistent volume
- ▶ Prevents log loss on Cloud Run container restarts / redeploys
- ▶ Unlocks cross-session reporting and audit trails

Cloud Scheduler for Daily Report

Planned

- ▶ Replace node-cron with Google Cloud Scheduler for reliability
- ▶ Eliminates cold-start risk — report fires even if container is idle
- ▶ Use existing /api/trigger-manager-report + x-admin-key pattern

Cache Pre-warm on Startup

Planned

- ▶ Build the customer search cache on server startup (not first request)
- ▶ Eliminates the slow first-open experience for customer search

MEDIUM TERM

GoFormz Integration

Planned

- ▶ Original scope from package name (simpro-goformz-automation)
- ▶ Auto-populate GoFormz inspection forms from simPRO job data
- ▶ Push completed form data back to simPRO as job notes / attachments

simPRO Webhook Integration

Planned

- ▶ Receive POST webhooks from simPRO when jobs are completed
- ▶ Trigger asset sync automatically (removes need for manual trigger)
- ▶ Enable real-time call sheet pre-population

MEDIUM TERM (CONT.)

Mobile-Optimised UI

Planned

- ▶ Responsive layouts for field technicians on phones/tablets
- ▶ Touch-friendly job detail cards and search modal
- ▶ Consider PWA (offline support via service worker)

Per-User Authentication

Deferred

- ▶ Replace shared ADMIN_API_KEY with individual user logins
- ▶ Options: Google OAuth (Workspace SSO) or JWT with email/pass
- ▶ Enable per-user activity tracking in email history

LONG TERM

Analytics Dashboard

Deferred

- ▶ Visualise call sheet volume, overdue site count, email trends
- ▶ Service date compliance tracking across the portfolio

Job Creation & Updates from UI

Deferred

- ▶ Currently read-only from simPRO — add write capabilities for jobs
- ▶ Allow creating follow-up jobs or updating job notes directly from FPOWS

MAINTENANCE SCHEDULE

Task	Frequency
Rotate SIMPRO_ACCESS_TOKEN	Per simPRO expiry policy
Rotate ADMIN_API_KEY	Quarterly
Review email_history.jsonl size	Monthly
Check Cloud Run logs for errors	Weekly
Verify daily report delivery	Weekly
Update Node.js dependencies	Quarterly